Methods for Proportions and Nominal Variables

I. Nominal (Categorical) Data
   A. The Population
   B. Estimating Proportions from Samples

II. Hypothesis Tests for Proportions
   A. *z* Test
   B. Yates's Continuity Correction

III. Examples
   A. Halothane vs. Morphine
   B. Aspirin vs. Placebo

IV Lecture Exercise

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	z Test Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5) Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.

†Not covered in this text.

‡Or interval data that are not necessarily normally distributed.

Categorical Data Comprise Much of Clinical Research

A. Nominal (no order) – e.g., Good Outcome: Y or N
B. Ordinal (ordered) – e.g., Pain Score: 0- no pain,…, 10- worst possible pain

Can Describe Events Involving Nominal Data by Counting and Calculating Proportions

Example: Halothane vs. Morphine in Heart Surgery

	Lived	Died	Total
Halothane	53	8	61
Morphine	57	10	67

Is this difference due to a clinical effect or just random sampling?

A. The Population

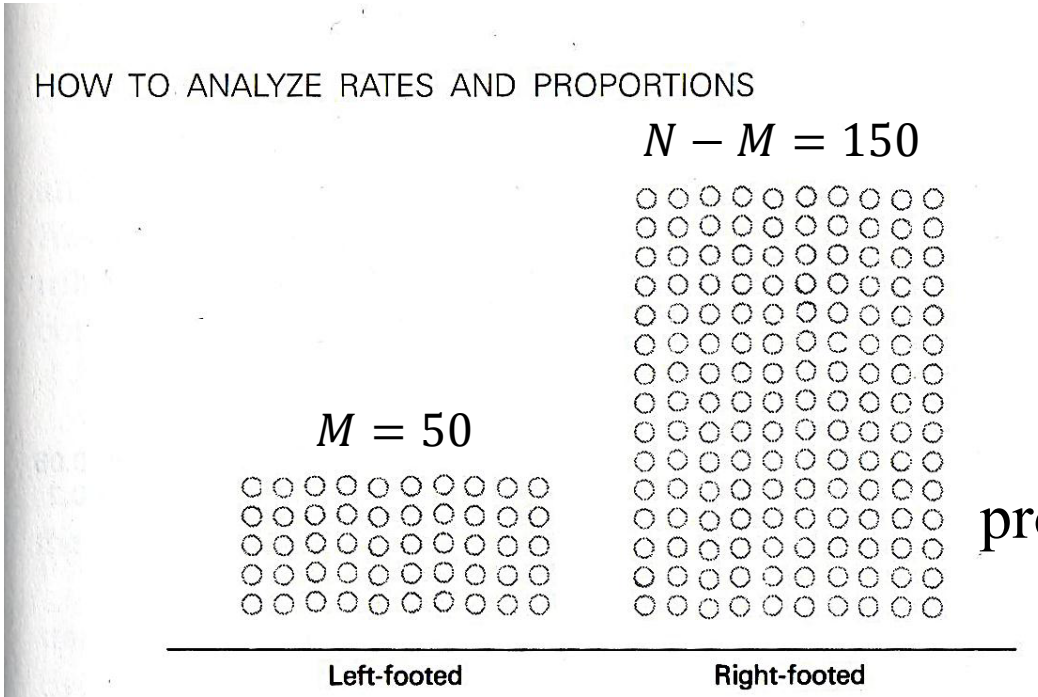


Figure 5-1 Of the 200 Martians 50 are left-footed, and the remaining 150 are right-footed. Therefore, if we select one Martian at random from this population, there is a $p_{\text{left}} = {}^{50}/_{200} = 0.25 = 25$ percent chance it will be left-footed.

$$N = 200$$

$$\text{prop. left} = \frac{M}{N} \equiv p$$

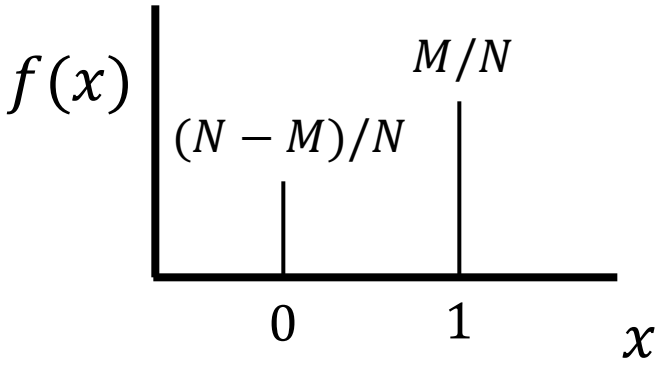
$$\text{prop. right} = \frac{N-M}{N} \equiv 1 - p$$

(left-footedness and right-footedness: two mutually exclusive and exhaustive groups)

In General

Given RV X (e.g., left-footedness, positive test result, etc.)

$x = 1$	(have attribute)	M
$x = 0$	(don't have attribute)	$N - M$



Mean of X

$x_i,\ i = 1, \rightleftharpoons, N$

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i = \frac{1}{N}\Big(M \cdot (1) + (N - M) \cdot (0) \Big) = \frac{M}{N}$$

Population Mean of a Proportion is called Population Proportion

$$p = \frac{M}{N}$$

Variance of X

$$x_i,\ i = 1, \ldots, N \qquad (x_i = 0 \ \text{ or } \ x_i = 1)$$

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \mu\right)^2 = E[X^2] - \left(\mu^2\right)$$

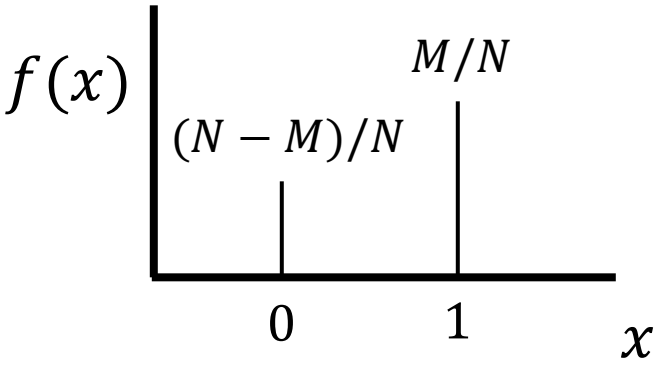
$$= \left(\frac{1}{N}\sum_{i=1}^{N} x_i^2\right) - \left(\mu^2\right) = \frac{M}{N} - \left(\frac{M}{N}\right)^2$$

$$= p - p^2 = p(1-p)$$

$$E[X^2] = \frac{1}{N}\sum_{i=1}^{N} {x_i}^2 = \frac{1}{N}(M \cdot 1^2 + (N-M)\cdot 0^2) = \frac{M}{N}$$

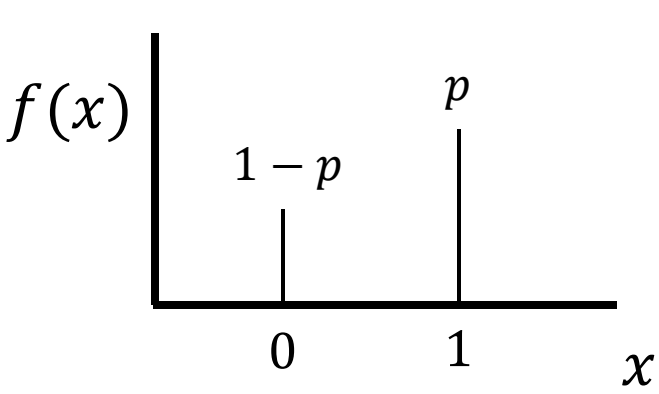
Population Variance

$$\sigma^2 = p(1-p)$$



$$p = \frac{M}{N}$$

More generally (using the p.f.)



Mean:

$$E[X] = \mu = \sum_{x \in D} xf(x)$$

$$\mu = \sum_{x \in D} xf(x) = 0(1-p) + 1p$$

$$\mu = p$$

Variance:

$$E[(X-p)^2] = \sigma^2 = \sum_{x \in D} (x-p)^2 f(x)$$

$$\sigma^2 = (0-p)^2(1-p) + (1-p)^2 p$$

$$\sigma^2 = p(1-p)$$

$$\sigma^2 = p(1-p)$$

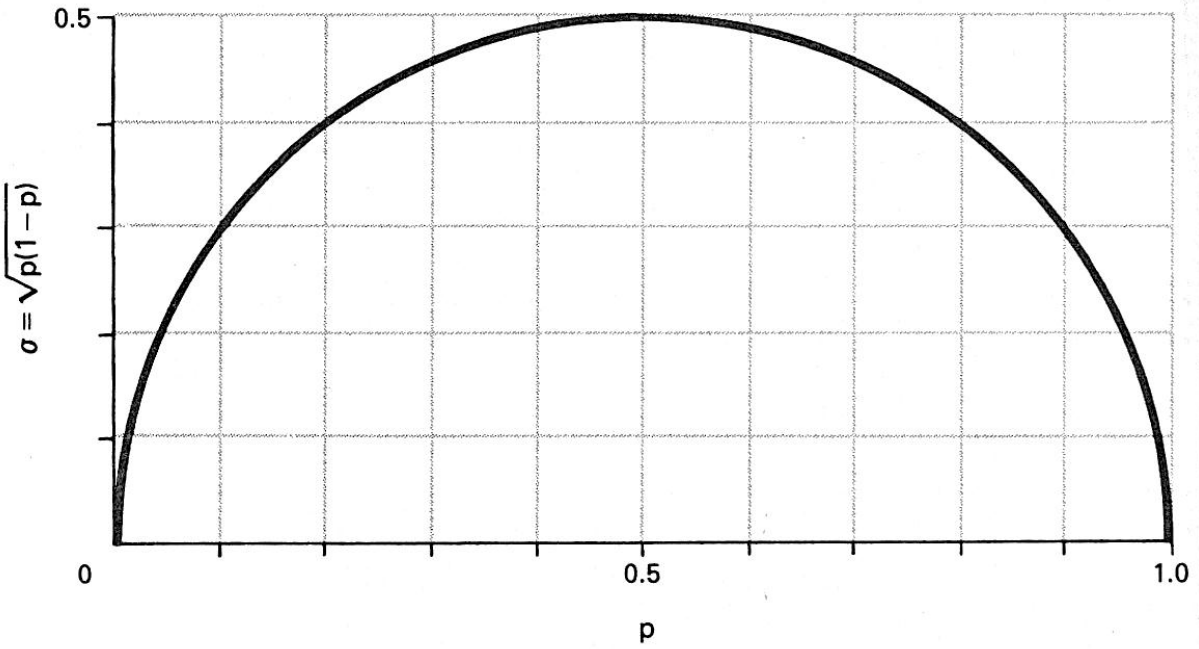


Figure 5–3 The relationship between the standard deviation of a population divided into two categories varies with p, the proportion of members in one of the categories. There is no variation if all members are in one category or the other (so $\sigma = 0$ when $p = 0$ or 1) and maximum variability when a given member is equally likely to fall in one class or the other ($\sigma = 0.5$ when $p = 0.5$).

B. Estimating Proportions from Samples

Sample of size 10

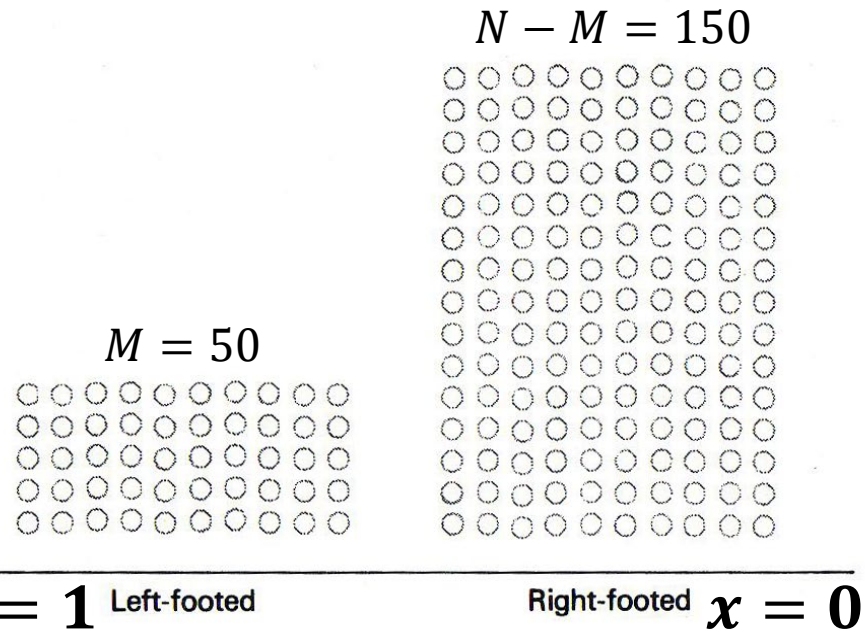


Figure 5-1 Of the 200 Martians 50 are left-footed, and the remaining 150 are right-footed. Therefore, if we select one Martian at random from this population, there is a $p_{left} = {}^{50}/_{200} = 0.25 = 25$ percent chance it will be left-footed.

B. Estimating Proportions from Samples

$x_i,\ i = 1,\ldots,n$ $\quad$ $x_i = 1$ in m subjects

sample proportion $\hat{p} = \frac{m}{n}$ ↓ Estimate of p

sample variance $s^2 = \hat{p}(1-\hat{p})$ ↓ Estimate of σ^2

B. Estimating Proportions from Samples

$x_i,\ i = 1,\ldots,n \qquad x_i = 1$ in m subjects

sample proportion $\hat{p} = \dfrac{m}{n}$

sample variance $s^2 = \hat{p}(1-\hat{p})$

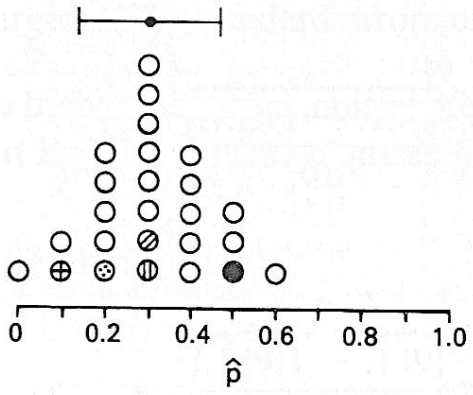


Figure 5-6 There will be a distribution of estimates of the proportion of left-footed Martians $\hat{p}_{left}$ depending on which random sample the investigator happens to draw. This figure shows the 5 specific random samples drawn in Figs. 5–4 and 5–5 together with 20 more random samples of 10 Martians each. The mean of the 25 estimates of p and the standard deviation of these estimates are also shown. The standard deviation of this distribution is the standard error of the estimate of the proportion $\sigma_{\hat{p}}$; it quantifies the precision with which $\hat{p}$ estimates p.

standard error of the proportion

$$s_{\hat{p}} = \sqrt{\frac{s^2}{n}}$$

$$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

B. Estimating Proportions from Samples

Example: Halothane vs. Morphine in Heart Surgery

	Lived	Died	Total
Halothane	53	8	61
Morphine	57	10	67

For Halothane (RV X represents death)

$$s_{\hat{p}} = 0.0432$$

$$\hat{p} = \frac{m}{n} = \frac{8}{61} = 0.131 \qquad s^2 = \hat{p}(1-\hat{p}) = 0.144 \quad \text{or} \quad s = 0.379$$

For Morphine (RV X represents death)

$$s_{\hat{p}} = 0.0435$$

$$\hat{p} = \frac{m}{n} = \frac{10}{67} = 0.149 \qquad s^2 = \hat{p}(1-\hat{p}) = 0.127 \quad \text{or} \quad s = 0.356$$

II. Hypothesis Test for Proportions

A. *z* Test

$$H_0: p_1 = p_2$$

$$\frac{\text{diff of sample proportions}}{\text{standard error of difference of sample proportions}} = \frac{\hat{p}_1 - \hat{p}_2}{s_{\hat{p}_1 - \hat{p}_2}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{s_{\hat{p}_1}^2 + s_{\hat{p}_2}^2}}$$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{s_{\hat{p}_1}^2 + s_{\hat{p}_2}^2}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \qquad z \sim N(0, 1)$$

Approximately

$$|z| > z_{critical}$$ Reject H_0

A. z Test

Group 1: n_1 subjects m_1 responses

Group 2: n_2 subjects m_2 responses

A better estimate

$$s_{\hat{p}_1-\hat{p}_2} = \sqrt{\frac{s^2}{n_1}+\frac{s^2}{n_2}} \qquad s = \sqrt{\hat{p}\left(1-\hat{p}\right)} \qquad \hat{p} = \frac{m_1+m_2}{n_1+n_2}$$

$$z = \frac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_1+1/n_2\right)}}$$

B. Yates-Corrected *z* Test

(always use Yates's Continuity Correction for 2x2 contingency tables)

$$z = \frac{\left|\hat{p}_1 - \hat{p}_2\right| - \frac{1}{2}\left(1/n_1 + 1/n_2\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_1 + 1/n_2\right)}} \qquad z \sim N(0, 1)$$

$$\hat{p}_1 = \frac{m_1}{n_1} \qquad \hat{p}_2 = \frac{m_2}{n_2} \qquad \hat{p} = \frac{m_1 + m_2}{n_1 + n_2}$$

$$z_{critical} = t_\alpha(\nu \to \infty) \qquad N(0, 1) = t(\nu \to \infty)$$

Using a two-tailed *t* table

$$\left|z\right| > z_{crit} = t_\alpha(\nu \to \infty) \text{ Reject } H_0 \qquad P < \alpha$$

III. Example: A. Halothane vs. Morphine in Heart Surgery

	Lived	Died	Total
Halothane (1)	53	8	61
Morphine (2)	57	10	67

$$H_0 : p_h = p_m$$

H0: There is no difference in the proportion of subjects who died when comparing the two anesthesia groups

III. Example: A. Halothane vs. Morphine in Heart Surgery

	Lived	Died	Total
Halothane (1)	53	8	61
Morphine (2)	57	10	67

$$H_0 : p_h = p_m$$

$$\hat{p}_h = \frac{m_h}{n_h} = \frac{8}{61} = 0.131 \qquad \hat{p}_m = \frac{m_m}{n_m} = \frac{10}{67} = 0.149 \qquad \hat{p} = \frac{m_h + m_m}{n_h + n_m} = 0.141$$

$$z = \frac{\left|\hat{p}_h - \hat{p}_m\right| - \frac{1}{2}\left(1/n_h + 1/n_m\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_h + 1/n_m\right)}} = 0.04$$

$$\left|z\right| = 0.04 \not> z_{crit,0.05} = t_{0.05}(\nu \to \infty) = 1.96$$ Cannot Reject H_0



Table 4–1 Critical Values of *t* (Two-Tailed)

	Probability of greater value, *P*								
ν	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
1	1.000	3.078	6.314	12.706	31.821	63.657	127.321	318.309	636.619
2	0.816	1.886	2.920	4.303	6.965	9.925	14.089	22.327	31.599
3	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.215	12.924
4	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.711	1.415	1.895	2.365	2.998	3.449	4.029	4.785	5.408
8	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768
24	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
31	0.682	1.309	1.696	2.040	2.453	2.744	3.022	3.375	3.633
32	0.682	1.309	1.694	2.037	2.449	2.738	3.015	3.365	3.622
33	0.682	1.308	1.692	2.035	2.445	2.733	3.008	3.356	3.611
34	0.682	1.307	1.691	2.032	2.441	2.728	3.002	3.348	3.601
35	0.682	1.306	1.690	2.030	2.438	2.724	2.996	3.340	3.591

(continued)

Table 4–1 Critical Values of *t* (Two-Tailed) (*Continued*)

	Probability of greater value, *P*								
ν	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
36	0.681	1.306	1.688	2.028	2.434	2.719	2.990	3.333	3.582
37	0.681	1.305	1.687	2.026	2.431	2.715	2.985	3.326	3.574
38	0.681	1.304	1.686	2.024	2.429	2.712	2.980	3.319	3.566
39	0.681	1.304	1.685	2.023	2.426	2.708	2.976	3.313	3.558
40	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
42	0.680	1.302	1.682	2.018	2.418	2.698	2.963	3.296	3.538
44	0.680	1.301	1.680	2.015	2.414	2.692	2.956	3.286	3.526
46	0.680	1.300	1.679	2.013	2.410	2.687	2.949	3.277	3.515
48	0.680	1.299	1.677	2.011	2.407	2.682	2.943	3.269	3.505
50	0.679	1.299	1.676	2.009	2.403	2.678	2.937	2.261	3.496
52	0.679	1.298	1.675	2.007	2.400	2.674	2.932	3.255	3.488
54	0.679	1.297	1.674	2.005	2.397	2.670	2.927	3.248	3.480
56	0.679	1.297	1.673	2.003	2.395	2.667	2.923	3.242	3.473
58	0.679	1.296	1.672	2.002	2.392	2.663	2.918	3.237	3.466
60	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
62	0.678	1.295	1.670	1.999	2.388	2.657	2.911	3.227	3.454
64	0.678	1.295	1.669	1.998	2.386	2.655	2.908	3.223	3.449
66	0.678	1.295	1.668	1.997	2.384	2.652	2.904	3.218	3.444
68	0.678	1.294	1.668	1.995	2.382	2.650	2.902	3.214	3.439
70	0.678	1.294	1.667	1.994	2.381	2.648	2.899	3.211	3.435
72	0.678	1.293	1.666	1.993	2.379	2.646	2.896	3.207	3.431
74	0.678	1.293	1.666	1.993	2.378	2.644	2.894	3.204	3.427
76	0.678	1.293	1.665	1.992	2.376	2.642	2.891	3.201	3.423
78	0.678	1.292	1.665	1.991	2.375	2.640	2.889	3.198	3.420
80	0.678	1.292	1.664	1.990	2.374	2.639	2.887	3.195	3.416
90	0.677	1.291	1.662	1.987	2.368	2.632	2.878	3.183	3.402
100	0.677	1.290	1.660	1.984	2.364	2.626	2.871	3.174	3.390
120	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
140	0.676	1.288	1.656	1.977	2.353	2.611	2.852	3.149	3.361
160	0.676	1.287	1.654	1.975	2.350	2.607	2.846	3.142	3.352
180	0.676	1.286	1.653	1.973	2.347	2.603	2.842	3.136	3.345
200	0.676	1.286	1.653	1.972	2.345	2.601	2.839	3.131	3.340
∞	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905
Normal	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905

Source: Adapted from J. H. Zar, *Biostatistical Analysis* (2 ed.), Prentice-Hall, Englewood Cliffs, N.J., 1984, pp. 484–485, table B.3. Used by permission.

III. Example: B. Effect of Aspirin on Thrombosis Formation

Table 5-1 Thrombus Formation in People Receiving Dialysis and Treated with Placebo or Aspirin

	Number of patients		
Sample group	Developed thrombi	Free of thrombi	Treated
Placebo	18	7	25
Aspirin	6	13	19
Total	24	20	44

Source: H. R. Harter, J. W. Burch, P. W. Majerus, N. Stanford, J. A. Delmez, C. B. Anderson, and C. A. Weerts "Prevention of Thrombosis in Patients on Hemodialysis by Low-Dose Aspirin," *N. Engl. J. Med.*, **301**:577–579, 1979. Reprinted by permission of the *New England Journal of Medicine.*

$$H_0 : p_{asp} = p_{pla}$$

that is, the proportion of subjects who develop thrombi is the same for the placebo and aspirin groups

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$$\hat{p}_{asp} = \frac{6}{19} = 0.316 \qquad \hat{p}_{pla} = \frac{18}{25} = 0.720 \qquad \hat{p} = \frac{6+18}{19+25} = 0.\overline{54}$$

$$z = \frac{\left|\hat{p}_{asp} - \hat{p}_{pla}\right| - \frac{1}{2}\left(1/n_{asp} + 1/n_{pla}\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_{asp} + 1/n_{pla}\right)}} = 2.36$$

Table 4–1 Critical Values of t (Two-Tailed) (*Continued*)

	Probability of greater value, P								
ν	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
Normal	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905

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Source: H. R. Harter, J. W. Burch, P. W. Majerus, N. Stanford, J. A. Delmez, C. B. Anderson, and C. A. Weerts "Prevention of Thrombosis in Patients on Hemodialysis by Low-Dose Aspirin," *N. Engl. J. Med.*, **301**:577–579, 1979. Reprinted by permission of the *New England Journal of Medicine.*

$$\hat{p}_{asp} = \frac{6}{19} = 0.316 \qquad \hat{p}_{pla} = \frac{18}{25} = 0.720 \qquad \hat{p} = \frac{6+18}{19+25} = 0.\overline{54}$$

$$z = \frac{\left|\hat{p}_{asp} - \hat{p}_{pla}\right| - \frac{1}{2}\left(1/n_{asp} + 1/n_{pla}\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_{asp} + 1/n_{pla}\right)}} = 2.36$$

$$|z| = 2.36 > t_{0.05}(\nu \to \infty) = 1.96$$

$$t_{0.02}(\nu \to \infty) = 2.3263$$

Reject H_0 $\quad P < 0.02$

Session 11 Lecture Exercise

5-1 Obtaining a blood sample of arterial blood permits measuring blood pH, oxygenation, and CO_2 elimination in order to see how well the lungs are functioning at oxygenating blood. The blood sample is often drawn from an artery in the wrist, which can be a painful procedure. Shawn Aaron and colleagues* compared the effectiveness of a topical anesthetic gel applied to the skin over the puncture point with a placebo cream. They observed adverse effects (redness, swelling, itching, or bruising) within 24 hours of administering the gel. Three of 36 people receiving the anesthetic gel and 8 of 40 receiving the placebo gel suffered an adverse reaction. Is there evidence of a difference in the rate of adverse effects between the anesthetic gel and the placebo gel?